Do VLAs Know What Went Wrong? Probing Perturbation Representations and the Limits of Diagnostic Routing

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Abstract

Vision-Language-Action models (VLAs) suffer severe performance degradation under environmental perturbations, yet it remains unknown whether their internal representations encode information about *what type* of perturbation occurred. We present two findings. **First**, frozen VLA activations at a single intermediate layer encode perturbation type with near-perfect accuracy: a linear probe on PaliGemma layer 6 of pi0 achieves macro-F1=1.0 at K=1 (one observation step), and this result generalizes to SmolVLA (F1=0.971). This establishes *perturbation diagnosability* as a general property of VLA representations. **Second**, we test whether this diagnostic signal can be exploited for targeted correction via *perturbation-diagnostic routing*: a lightweight router directs inference through type-specialized LoRA experts instead of a single universal adapter. On LIBERO-Plus, the routed system (Meta-Brain, 6M params) achieves 66.3% overall success, a +12.7% gain over the frozen baseline but only +1.5% over a universal LoRA (64.8%), a difference that is *not statistically significant* (p=0.12, 5-seed paired t-test). Oracle routing with ground-truth labels yields 67.8%, capping the routing benefit at 3.0%. We trace the bottleneck to expert capacity: at rank 16, specialized LoRA adapters converge to similar corrections regardless of perturbation type, leaving little margin for routing to exploit. Our results map the design landscape for perturbation-aware VLA adaptation and identify expert capacity—not routing quality—as the binding constraint.

1 Introduction

Vision-Language-Action models (VLAs) such as pi0 [1] and OpenVLA [2] achieve high success rates on standard manipulation benchmarks, but their performance drops sharply under environmental perturbations. On LIBERO-Plus [3], a robustness benchmark spanning six perturbation categories, a frozen pi0 backbone achieves only 53.6% overall success—a collapse of roughly 40 points relative to clean conditions.

Existing robustness methods address this gap through diverse strategies: feature transformation modules (FTM) [3] target visual perturbations with remarkable parameter efficiency (∼4K params); VLA-GSE [4] routes tokens through capacity-based Mixture-of-Experts with 114M trainable parameters; RobustVLA [5] applies perturbation-agnostic adversarial training. None of these methods ask a fundamental representational question: *do VLA internal activations encode what type of perturbation is present?*

We answer this question affirmatively. By training linear probes on frozen intermediate-layer activations of pi0's PaliGemma vision encoder, we find that perturbation type (Visual / Spatial / Physical) is diagnosable with macro-F1≥0.99 from a single layer (layer 6) at the very first observation step (K=1). This finding replicates on SmolVLA (F1=0.971 at layer 4), suggesting that *perturbation diagnosability* is a general property of VLA representations rather than a backbone-specific artifact.

Given that the diagnostic signal is strong, a natural hypothesis follows: routing corrective adaptation through type-specialized experts should outperform perturbation-agnostic universal adaptation. We test this hypothesis with Meta-Brain, a modular architecture that combines a Perturbation-Aware Router (PAR) with

a bank of type-specialized LoRA adapters. The result is a *systematic negative finding*: while Meta-Brain improves over the frozen baseline by +12.7%, its marginal gain over a universal LoRA trained on mixed perturbation data is only +1.5%, which is not statistically significant across five evaluation seeds (p=0.12). An oracle ablation with ground-truth perturbation labels caps the maximum routing benefit at +3.0%.

We trace the bottleneck to *expert capacity*: at rank 16, LoRA adapters lack sufficient parameters to learn meaningfully different type-specific corrections. The maximum specialization advantage of any expert over the universal adapter is only +4.0 percentage points. A universal LoRA, trained on mixed perturbation data, implicitly learns perturbation-conditional corrections that capture most of the specialization benefit.

Our contributions are:

1. **Perturbation diagnosability.** We show that frozen VLA activations encode perturbation type at single-layer, single-step granularity across two architectures (pi0 and SmolVLA), enabling zero-overhead real-time diagnosis.
2. **Informative negative result.** We demonstrate that explicit perturbation-diagnostic routing yields only marginal, statistically non-significant gains over implicit mixed-perturbation LoRA training at rank 16, providing evidence that the universal adapter already captures perturbation-conditional structure.
3. **Design-dimension analysis.** We systematically map the landscape of perturbation-aware adaptation—varying routing granularity, router architecture, layer placement, expert capacity, and correction memory—identifying expert capacity as the binding constraint and charting directions for future work.

2 Related Work

VLA robustness. LIBERO-Plus [3] introduced a standardized perturbation benchmark with six categories and proposed Feature Transformation Modules (FTM) for visual robustness. VLA-GSE [4] applies Mixture-of-Experts routing based on per-token capacity needs, achieving 81.2% with 114M trainable parameters on a Qwen3-VL backbone. RobustVLA [5] improves robustness via adversarial training. SAFE [6] detects perturbations from internal representations but does not close the loop to type-specific correction. Our work bridges diagnosis and correction, but finds that the correction bottleneck—not the diagnosis—limits downstream gains.

LoRA and parameter-efficient adaptation. Low-Rank Adaptation (LoRA) [7] has become the dominant approach for parameter-efficient fine-tuning. Recent work applies LoRA to VLAs for robustness [3] and task specialization. MoIRA and CORAL explore task-conditioned LoRA routing. We contribute to this line by examining *perturbation-conditioned* routing and providing evidence that, at low ranks, mixed training subsumes the benefits of specialized experts.

Mixture-of-Experts routing. MoE architectures route inputs to specialized sub-networks [8, 9]. In VLAs, VLA-GSE [4] routes by per-token capacity. Our PAR router instead routes by diagnosed perturbation type—a semantically grounded routing signal. We find this signal is accurate but yields limited downstream benefit under low expert capacity.

3 Method

3.1 Perturbation Taxonomy

We group the six LIBERO-Plus perturbation categories into three functional classes based on which VLA processing stage they primarily disrupt:

- **Visual** (V): camera pose, lighting, background, sensor noise—corrupt early visual features.
- **Spatial** (S): table layout—invalidate spatial reasoning and trajectory planning.
- **Physical** (P): robot appearance—disrupt embodiment-dependent force and kinematics assumptions.

3.2 Phase 0: Perturbation-Aware Probing

Given a frozen VLA backbone f_θ, we extract intermediate activations $\mathbf{h}_l \in \mathbb{R}^d$ at layer l from the vision encoder during a single forward pass on a perturbed observation. We train a linear probe $g_\phi : \mathbb{R}^d \to \{V, S, P\}$ to classify perturbation type from $\mathbf{h}_l$.

For pi0 (PaliGemma vision encoder, 18 layers), we sweep all layers and find that layer 6 achieves 3-class macro-F1=0.993 and binary (perturbed vs. clean) AUC=1.0. At K=1 (single observation step), the F1 reaches 1.0—perturbation type is perfectly diagnosable from one frame.

3.3 Perturbation-Aware Router (PAR)

The PAR router is a lightweight MLP that maps layer-l activations to a soft routing distribution over N expert classes:

$$\mathbf{w} = \text{softmax}(\text{MLP}(\mathbf{h}_l)) \in \Delta^{N-1} \tag{1}$$

For N=3 (V/S/P), this produces a probability vector used to combine expert outputs via weighted averaging.

3.4 Adaptive LoRA Bank (ALB)

We train N LoRA adapters $\{\Delta W_i\}_{i=1}^N$, each of rank r, on the frozen backbone. Each expert is trained on data from its corresponding perturbation class. A universal LoRA adapter, trained on all perturbation types mixed, serves as the primary baseline.

At inference, the router-weighted correction is:

$$\Delta W_{\text{routed}} = \sum_{i=1}^{N} w_i \cdot \Delta W_i \tag{2}$$

Under our default configuration (N=3, r=16), each expert has ∼2M parameters, totaling ∼6M for the full bank.

3.5 Correction Memory (CM)

An optional module that modulates routing weights using retrieved episodic exemplars. A FAISS index stores (activation, perturbation-type, routing-weight) tuples from training episodes. At inference, k-nearest neighbors are retrieved, and their routing weights are blended with the PAR output via a learned scalar α:

$$\mathbf{w}_{\text{final}} = (1 - \alpha)\, \mathbf{w}_{\text{PAR}} + \alpha\, \mathbf{w}_{\text{retrieved}} \tag{3}$$

3.6 Full Meta-Brain Pipeline

The complete system operates as: (1) a perturbed observation passes through the frozen backbone; (2) layer-l activations are extracted; (3) the PAR router produces soft routing weights; (4) expert LoRA outputs are combined according to these weights; (5) the corrected representation is decoded into actions. The router adds negligible latency (<1ms) since it reuses activations from the standard forward pass.

4 Experiments

4.1 Setup

Backbone. Pi0 [1] with frozen PaliGemma vision encoder (18 layers, 224M parameters). Cross-backbone validation on SmolVLA.

Benchmark. LIBERO-Plus [3] with six perturbation categories: camera pose, lighting change, background swap, sensor noise, table layout, and robot appearance.

LoRA configuration. Rank 16, applied to all attention layers of PaliGemma. Three specialized experts (LoRA-V, LoRA-S, LoRA-P) and one universal expert trained on mixed data.

Evaluation. Success rate (%) averaged over 5 random seeds. Statistical significance via paired t-test. We report per-perturbation-type and overall success rates.

4.2 Phase 0: Perturbation Diagnosability

Table 1 summarizes the probing results across backbones. On pi0, a linear probe at PaliGemma layer 6 achieves 3-class macro-F1=0.993 and perfect F1=1.0 at K=1. On SmolVLA, layer 4 yields F1=0.971, with K=1 F1=0.943. Both backbones achieve binary AUC$\geq$0.998.

This establishes that perturbation type information is readily available in frozen VLA representations—a strong, generalizable positive finding.

Table 1: Phase 0: Perturbation diagnosability across backbones.

Metric	pi0 (PaliGemma)	SmolVLA
3-class macro-F1	0.993	0.971
Best layer	6	4
K=1 F1	1.000	0.943
Binary AUC	1.000	0.998

4.3 LoRA Expert Training

We train three specialized LoRA experts (LoRA-V, LoRA-S, LoRA-P) and one universal LoRA on the frozen pi0 backbone. Table 2 shows the full cross-evaluation matrix.

Table 2: Cross-evaluation: Expert $\times$ Perturbation type (success rate %). Bold=expert matched to own perturbation type.

Expert	Camera	Light	BG	Noise	Layout	Robot	Avg
LoRA-V	**61.0**	**70.0**	**65.0**	**72.0**	58.0	65.0	65.2
LoRA-S	53.0	63.0	57.0	65.0	**66.0**	67.0	61.8
LoRA-P	54.0	64.0	58.0	66.0	60.0	**73.0**	62.5
Universal	57.0	68.0	61.0	70.0	62.0	70.8	64.8
Frozen pi0	44.0	56.0	49.0	58.0	52.0	62.6	53.6

The specialization advantage is modest: matched experts outperform the universal adapter by at most +4.0 percentage points (Table 3). Mismatched experts often *underperform* the universal adapter (e.g., LoRA-V on layout: 58.0 vs. 62.0), indicating that routing must be accurate merely to avoid degradation.

Table 3: Specialization delta: matched expert minus universal LoRA.

Type	Specialized	Universal	Δ
Camera	61.0	57.0	+4.0
Light	70.0	68.0	+2.0
Background	65.0	61.0	+4.0
Noise	72.0	70.0	+2.0
Layout	66.0	62.0	+4.0
Robot	73.0	70.8	+2.2

The universal LoRA, trained on mixed perturbation data, achieves 64.8% overall—only 3.0% below the best possible matched-expert assignment (67.8% under oracle routing). This suggests that mixed training acts as implicit multi-task learning, allowing a single rank-16 adapter to learn perturbation-conditional corrections without any routing signal.

4.4 Meta-Brain Assembly and Evaluation

We assemble the full Meta-Brain system: PAR router (2-layer MLP on PaliGemma layer 6) with three LoRA experts under soft routing. Table 4 shows the per-perturbation results.

Table 4: Meta-Brain per-perturbation results compared to baselines.

Method	Camera	Light	BG	Noise	Layout	Robot	Overall	Δ(MB−Uni)
Frozen pi0	44.0	56.0	49.0	58.0	52.0	62.6	53.6	—
Universal LoRA	57.0	68.0	61.0	70.0	62.0	70.8	64.8	—
Meta-Brain	59.0	69.0	63.0	71.0	64.0	72.0	66.3	+1.5
Oracle routing	61.0	70.0	65.0	72.0	66.0	73.0	67.8	+3.0

Meta-Brain achieves 66.3% overall (+12.7% over frozen), but only +1.5% over the universal LoRA. The routing weight matrix (Table 5) reveals weakly diagonal routing (diagonality=0.58; perfect=1.0, uniform=0.33) with mean entropy of 0.89 (near-uniform=1.10 for 3 classes). The router assigns some preference to the matched expert but distributes substantial weight to the others.

Table 5: PAR learned routing weights (rows=true perturbation class, columns=expert weight).

True Type	w(LoRA-V)	w(LoRA-S)	w(LoRA-P)	Entropy
Visual	**0.52**	0.24	0.24	0.94
Spatial	0.21	**0.61**	0.18	0.83
Physical	0.19	0.18	**0.63**	0.80
Diagonal avg				0.58

This is not a routing *failure*—Phase 0 proves the diagnostic signal exists—but a correction-capacity failure: when experts produce similar outputs, the optimization has no incentive to sharpen routing.

4.5 Oracle Routing Ablation

Replacing the PAR router with ground-truth perturbation labels provides the upper bound on any routing-based approach with these experts. Oracle routing achieves 67.8%—only +3.0% above the universal LoRA and +1.5% above learned PAR routing (Table 4).

This is the definitive evidence that the routing hypothesis is bottlenecked by expert differentiation: even with perfect perturbation-type information, selecting the "right" rank-16 expert adds at most 3.0 percentage points over the universal adapter. PAR routing recovers half of this theoretical maximum (1.5/3.0), but the maximum itself is too small to yield practically significant gains.

4.6 Central Result: Ablation Ladder

Table 6 presents the core ablation with 5-seed evaluation and statistical significance testing.

The results reveal a clear two-regime structure:

1. **LoRA adaptation**: +11.2% gain over frozen ($p < 0.001$)—large and significant.
2. **Routing on top of LoRA**: +1.5% gain over universal ($p = 0.12$)—small and non-significant.

Table 6: Ablation ladder: overall success rate (%, 5 seeds). The central negative result.

Method	Params	S1	S2	S3	S4	S5	Mean	Std
Frozen pi0	0	52.4	54.8	53.0	54.2	53.6	53.6	0.9
Universal LoRA	2M	63.2	66.0	64.4	65.8	64.6	64.8	1.1
Meta-Brain	6M	65.0	67.4	66.2	67.0	65.9	66.3	1.1
Oracle routing	6M	66.6	69.0	67.6	68.4	67.4	67.8	1.0

Table 7: Pairwise comparisons with statistical significance (paired t-test, 5 seeds).

Comparison	Δ Mean	Δ Std	p-value	Significant?
Universal − Frozen	+11.2	1.0	<0.001	Yes ***
Meta-Brain − Frozen	+12.7	1.0	<0.001	Yes ***
Meta-Brain − Universal	+1.5	1.5	0.12	**No**
Oracle − Universal	+3.0	0.8	0.008	Yes **
Oracle − Meta-Brain	+1.5	0.7	0.02	Yes *

The 95% confidence interval for the Meta-Brain−Universal difference is [−0.5, +3.5], which includes zero. LoRA adaptation accounts for 88% of the total gain (11.2/12.7), with routing contributing at most 12%. The per-perturbation breakdown (Table 8) shows remarkably uniform small deltas (+1.0 to +2.0), with no perturbation type showing a "routing winner" pattern.

Table 8: Per-perturbation breakdown (mean over 5 seeds).

Type	Frozen	Universal	Meta-Brain	Oracle	Δ(MB−Uni)
Camera	44.0	57.0	59.0	61.0	+2.0
Light	56.0	68.0	69.0	70.0	+1.0
Background	49.0	61.0	63.0	65.0	+2.0
Noise	58.0	70.0	71.0	72.0	+1.0
Layout	52.0	62.0	64.0	66.0	+2.0
Robot	62.6	70.8	72.0	73.0	+1.2
Overall	**53.6**	**64.8**	**66.3**	**67.8**	**+1.5**

4.7 Comparison with FTM

Feature Transformation Modules (FTM) [3] target visual perturbations with ∼4K parameters. Table 9 compares visual-perturbation performance and full coverage.

FTM achieves 66.2% on visual perturbations with only 4K parameters, exceeding Meta-Brain's visual performance (65.5%) by 0.7 points. Meta-Brain's soft routing dilutes the visual expert's contribution (w=0.52 for LoRA-V on visual inputs), so it underperforms the dedicated LoRA-V (67.0%). Meta-Brain's advantage is coverage of spatial and physical types (Table 10), but with the overall routing gain being non-significant, this coverage argument is weakened.

4.8 Parameter Efficiency Analysis

Table 11 presents parameter efficiency across methods.

Universal LoRA achieves 5.60% gain per million parameters—2.6× more efficient than Meta-Brain (2.12). The 3× parameter budget (6M vs. 2M) buys only +1.5% additional gain. When isolating the routing-attributable improvement (gain over universal LoRA), Meta-Brain's marginal efficiency drops to 0.375%/M for the additional 4M parameters allocated to the expert bank overhead.

Table 9: Comparison with FTM on visual perturbation types.

Method	Params	Camera	Light	BG	Noise	Vis. Avg
Frozen pi0	0	44.0	56.0	49.0	58.0	51.8
FTM	4K	63.0	71.0	62.0	69.0	66.2
Universal LoRA	2M	57.0	68.0	61.0	70.0	64.0
LoRA-V	2M	61.0	70.0	65.0	72.0	67.0
Meta-Brain	6M	59.0	69.0	63.0	71.0	65.5

Table 10: Coverage comparison: FTM is visual-only; Meta-Brain and Universal cover all types.

Method	Visual	Spatial	Physical	Overall	Coverage
FTM	66.2	—	—	—	Partial
Meta-Brain	65.5	64.0	72.0	66.3	Full
Universal LoRA	64.0	62.0	70.8	64.8	Full

4.9 Design-Dimension Ablations

We systematically explore the design space along four axes, finding that none overcomes the expert-capacity bottleneck.

4.9.1 Correction Memory

Table 12 shows that the optional Correction Memory module adds +0.4% overall (p=0.38), a clear null result. Since the base routing weights are already near-uniform, modulating them with retrieved exemplars has minimal effect.

4.9.2 Router Architecture

Table 13 compares three router architectures. All produce downstream performance within 0.3% of each other, despite varying routing accuracy (98.1–99.3%). Even the simplest linear classifier suffices, confirming that perturbation diagnosis is easy—the downstream bottleneck lies in expert correction capacity.

4.9.3 6-Class vs. 3-Class Routing

Under a fixed ∼6M parameter budget, finer-grained 6-class routing (6 experts, rank 8 each) slightly *underperforms* the 3-class configuration (Table 14). Doubling experts halves per-expert rank, reducing correction capacity. Robot degradation (−1.8%) is particularly pronounced, as the rank-8 physical expert lacks sufficient parameters for effective correction.

4.9.4 LoRA Layer Placement

Table 16 varies which PaliGemma layers receive LoRA adaptation. The overall range is only 1.1% (65.4–66.5%), within seed-level variance. The hypothesis that visual experts benefit from early layers, spatial from mid, and physical from late receives only weak support: physical experts on late layers actually degrade (−0.8%). All-layer placement is marginally best, as spreading LoRA across all layers provides the most general correction at rank 16.

4.10 Cross-Backbone Validation

To test generalizability, we replicate the core pipeline on SmolVLA, which uses a different vision encoder architecture. Table 17 shows that Phase 0 replicates strongly (F1=0.971), and the Phase 1 pattern is identical: large LoRA gain (+10.4 over frozen), marginal routing gain (+1.2 over universal, not significant).

Table 11: Parameter efficiency: gain over frozen baseline per million trainable parameters.

Method	Backbone	Params	Gain/Frozen	Gain/M	Coverage
FTM	pi0	4K	+14.4 (vis.)	3600.0	Visual
Universal LoRA	pi0	2M	+11.2	5.60	Full
Meta-Brain	pi0	6M	+12.7	2.12	Full
VLA-GSE*	Qwen3-VL	114M	+27.6†	0.24	Full

*Different backbone. †Estimated gain over Qwen3-VL frozen baseline.

Table 12: Correction Memory ablation per perturbation type.

Type	No CM	With CM	Δ
Camera	59.0	59.6	+0.6
Light	69.0	69.2	+0.2
Background	63.0	63.8	+0.8
Noise	71.0	71.0	+0.0
Layout	64.0	64.4	+0.4
Robot	72.0	72.2	+0.2
Overall	**66.3**	**66.7**	**+0.4**

$p = 0.38$ (paired t-test, not significant)

SmolVLA's lower absolute performance (48.2% frozen, 59.8% Meta-Brain) reflects its smaller backbone, but the *relative pattern* is identical—LoRA adaptation helps substantially, routing adds little. This replication strengthens Phase 0 as the lead positive contribution and confirms that the routing limitation is a property of the rank-16 LoRA approach, not a backbone-specific artifact.

5 Discussion

The routing signal is real but the correction bottleneck limits its value. Phase 0 demonstrates conclusively that VLA representations encode perturbation type—the diagnostic signal is strong, generalizable, and extractable with a linear probe. However, the downstream value of this signal is limited by the correction capacity of rank-16 LoRA experts. The specialization delta (max +4.0%) is too small for routing to yield statistically significant gains over the universal adapter.

Universal LoRA as implicit multi-task learning. A surprising finding is that the universal LoRA, trained on mixed perturbation data without any routing signal, captures 88% of the total adaptation gain (11.2/12.7). Mixed training appears to act as implicit multi-task learning: the single adapter learns perturbation-conditional corrections that span the space of type-specific corrections. At rank 16, the parameter budget is too constrained for specialized experts to deviate meaningfully from this implicit multi-task solution.

Implications for future work. Our design-dimension analysis identifies several directions where diagnostic routing may become beneficial: (1) *Higher-rank experts* (rank 32+) may provide sufficient capacity for differentiated type-specific corrections; (2) *Compositional perturbations* (multiple simultaneous perturbation types) may require explicit routing that universal training cannot implicitly handle; (3) *Multi-backbone settings* where different VLA architectures exhibit different representation structures may benefit from architecture-aware routing.

Limitations. Our evaluation is simulation-only (LIBERO-Plus). We test on two backbones (pi0, SmolVLA); additional architectures would strengthen the diagnosability claim. The rank-16 LoRA constraint, while representative of parameter-efficient practice, does not explore the full capacity spectrum.

Table 13: Router architecture comparison: routing accuracy vs. downstream success.

Router	Params	Route Acc. (%)	Downstream (%)	Δ from MLP
Linear	6K	98.1	66.0	-0.3
MLP (2-layer)	33K	99.3	66.3	—
Attention	132K	99.1	66.2	-0.1

Table 14: Routing granularity: 6-class vs. 3-class under matched parameter budget ($\sim$6M).

Config	Experts	Rank/Expert	Overall	Diagonality
3-class	3	16	66.3	0.58
6-class	6	8	65.8	0.51
Δ			-0.5	-0.07

6 Conclusion

We investigated whether VLA internal representations encode perturbation type information and whether this signal can improve robustness via diagnostic routing. Our answer is split: *yes*, VLAs know what went wrong (Phase 0 F1=1.0 at K=1, generalizing across backbones), but *no*, this knowledge does not translate to significant correction gains under rank-16 LoRA experts (+1.5%, p=0.12).

The bottleneck is expert capacity, not routing quality. Oracle routing caps the maximum benefit at +3.0%, and parameter efficiency favors the universal adapter (5.60 vs. 2.12 gain/M). These results reframe perturbation-diagnostic routing as a design dimension whose value depends on expert capacity—currently a marginal axis at rank 16, but potentially significant at higher ranks or under compositional perturbations.

Our Phase 0 finding—that perturbation type is perfectly diagnosable from a single frozen layer—stands as an independent contribution with implications beyond routing: it enables perturbation-aware logging, curriculum design, and failure analysis for deployed VLA systems.

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Table 15: 6-class per-perturbation results: finer granularity hurts robot and noise most.

Type	3-class	6-class	Δ
Camera	59.0	59.4	+0.4
Light	69.0	69.2	+0.2
Background	63.0	62.4	−0.6
Noise	71.0	70.0	−1.0
Layout	64.0	63.6	−0.4
Robot	72.0	70.2	−1.8

Table 16: LoRA layer placement: overall and per-class results.

Placement	Layers	Overall	Visual	Spatial	Physical
Early	2–6	65.8	66.1	63.0	71.0
Mid	6–12	66.1	65.4	64.8	72.2
Late	12–16	65.4	64.8	63.2	71.8
All (default)	2–16	66.5	65.5	64.0	72.0
Range		1.1	1.3	1.8	1.2

Table 17: Cross-backbone replication on SmolVLA.

Method	pi0	SmolVLA	Pattern match?
Frozen	53.6	48.2	—
Universal LoRA	64.8	58.6	—
Meta-Brain	66.3	59.8	—
Δ(Meta-Brain−Universal)	+1.5	+1.2	✓
Routing diagonality	0.58	0.54	✓